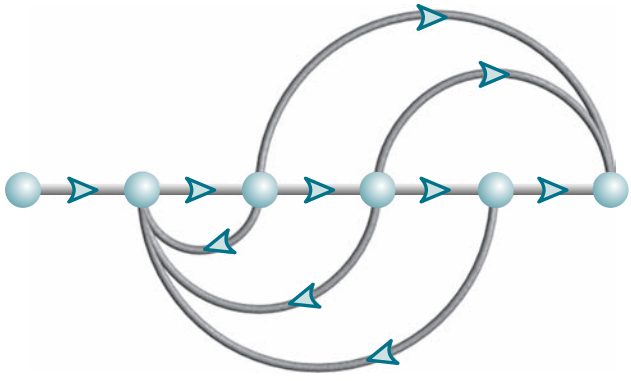
Stability

6



Chapter Learning Outcomes

After completing this chapter the student will be able to:

- Make and interpret a basic Routh table to determine the stability of a system (Sections 6.1–6.2)
- Make and interpret a Routh table where either the first element of a row is zero or an entire row is zero (Sections 6.3–6.4)
- Use a Routh table to determine the stability of a system represented in state space (Section 6.5)

State Space
SS

Case Study Learning Outcomes

You will be able to demonstrate your knowledge of the chapter objectives with case studies as follows:

- Given the antenna azimuth position control system shown on the front endpapers, you will be able to find the range of preamplifier gain to keep the system stable.
- Given the block diagrams for the UFSS vehicle's pitch and heading control systems on the back endpapers, you will be able to determine the range of gain for stability of the pitch or heading control system.

6.1 Introduction

In Chapter 1, we saw that three requirements enter into the design of a control system: transient response, stability, and steady-state errors. Thus far we have covered transient response, which we will revisit in Chapter 8. We are now ready to discuss the next requirement, stability.

Stability is the most important system specification. If a system is unstable, transient response and steady-state errors are moot points. An unstable system cannot be designed for a specific transient response or steady-state error requirement. What, then, is stability? There are many definitions for stability, depending upon the kind of system or the point of view. In this section, we limit ourselves to linear, time-invariant systems.

In Section 1.5, we discussed that we can control the output of a system if the steady-state response consists of only the forced response. But the total response of a system is the sum of the forced and natural responses, or

$$c(t) = c_{\text{forced}}(t) + c_{\text{natural}}(t) \tag{6.1}$$

Using these concepts, we present the following definitions of stability, instability, and marginal stability:

A linear, time-invariant system is *stable* if the natural response approaches zero as time approaches infinity.

A linear, time-invariant system is *unstable* if the natural response grows without bound as time approaches infinity.

A linear, time-invariant system is *marginally stable* if the natural response neither decays nor grows but remains constant or oscillates as time approaches infinity.

Thus, the definition of stability implies that only the forced response remains as the natural response approaches zero.

These definitions rely on a description of the natural response. When one is looking at the total response, it may be difficult to separate the natural response from the forced response. However, we realize that if the input is bounded and the total response is not approaching infinity as time approaches infinity, then the natural response is obviously not approaching infinity. If the input is unbounded, we see an unbounded total response, and we cannot arrive at any conclusion about the stability of the system; we cannot tell whether the total response is unbounded because the forced response is unbounded or because the natural response is unbounded. Thus, our alternate definition of *stability*, one that regards the total response and implies the first definition based upon the natural response, is this:

A system is stable if *every* bounded input yields a bounded output.

We call this statement the bounded-input, bounded-output (BIBO) definition of stability.

Let us now produce an alternate definition for instability based on the total response rather than the natural response. We realize that if the input is bounded but the total response is unbounded, the system is unstable, since we can conclude that the natural response approaches infinity as time approaches infinity. If the input is unbounded, we will see an unbounded total response, and we cannot draw any conclusion about the stability of the system; we cannot tell whether the total response is unbounded because the forced response is unbounded or because the

natural response is unbounded. Thus, our alternate definition of *instability*, one that regards the total response, is this:

A system is unstable if *any* bounded input yields an unbounded output.

These definitions help clarify our previous definition of *marginal stability*, which really means that the system is stable for some bounded inputs and unstable for others. For example, we will show that if the natural response is undamped, a bounded sinusoidal input of the same frequency yields a natural response of growing oscillations. Hence, the system appears stable for all bounded inputs except this one sinusoid. Thus, marginally stable systems by the natural response definitions are included as unstable systems under the BIBO definitions.

Let us summarize our definitions of stability for linear, time-invariant systems. Using the natural response:

1. A system is stable if the natural response approaches zero as time approaches infinity.
2. A system is unstable if the natural response approaches infinity as time approaches infinity.
3. A system is marginally stable if the natural response neither decays nor grows but remains constant or oscillates.

Using the total response (BIBO):

1. A system is stable if *every* bounded input yields a bounded output.
2. A system is unstable if *any* bounded input yields an unbounded output.

Physically, an unstable system whose natural response grows without bound can cause damage to the system, to adjacent property, or to human life. Many times systems are designed with limited stops to prevent total runaway. From the perspective of the time response plot of a physical system, instability is displayed by transients that grow without bound and, consequently, a total response that does not approach a steady-state value or other forced response.[1]

How do we determine if a system is stable? Let us focus on the natural response definitions of stability. Recall from our study of system poles that poles in the left half-plane (lhp) yield either pure exponential decay or damped sinusoidal natural responses. These natural responses decay to zero as time approaches infinity. Thus, if the closed-loop system poles are in the left half of the plane and hence have a negative real part, the system is stable. That is, *stable systems have closed-loop transfer functions with poles only in the left half-plane.*

Poles in the right half-plane (rhp) yield either pure exponentially increasing or exponentially increasing sinusoidal natural responses. These natural responses approach infinity as time approaches infinity. Thus, if the closed-loop system poles are in the right half of the s-plane and hence have a positive real part, the system is unstable. Also, poles of multiplicity greater than 1 on the imaginary axis lead to the sum of responses of the form $At^n \cos(\omega t + \phi)$, where $n = 1, 2, \ldots$, which also approaches infinity as time approaches infinity. Thus, *unstable systems have closed-loop transfer functions with at least one pole in the right half-plane and/or poles of multiplicity greater than 1 on the imaginary axis.*

[1] Care must be taken here to distinguish between natural responses growing without bound and a forced response, such as a ramp or exponential increase, that also grows without bound. A system whose forced response approaches infinity is stable as long as the natural response approaches zero.

Finally, a system that has imaginary axis poles of multiplicity 1 yields pure sinusoidal oscillations as a natural response. These responses neither increase nor decrease in amplitude. Thus, *marginally stable systems have closed-loop transfer functions with only imaginary axis poles of multiplicity 1 and poles in the left half-plane.*

As an example, the unit step response of the stable system of Figure 6.1(*a*) is compared to that of the unstable system of Figure 6.1(*b*). The responses, also shown in Figure 6.1, show that while the oscillations for the stable system diminish, those for the unstable system increase without bound. Also notice that the stable system's response in this case approaches a steady-state value of unity.

It is not always a simple matter to determine if a feedback control system is stable. Unfortunately, a typical problem that arises is shown in Figure 6.2. Although we know the poles of the forward transfer function in Figure 6.2(*a*), we do not know the location of the poles of the equivalent closed-loop system of Figure 6.2(*b*) without factoring or otherwise solving for the roots.

However, under certain conditions, we can draw some conclusions about the stability of the system. First, if the closed-loop transfer function has only

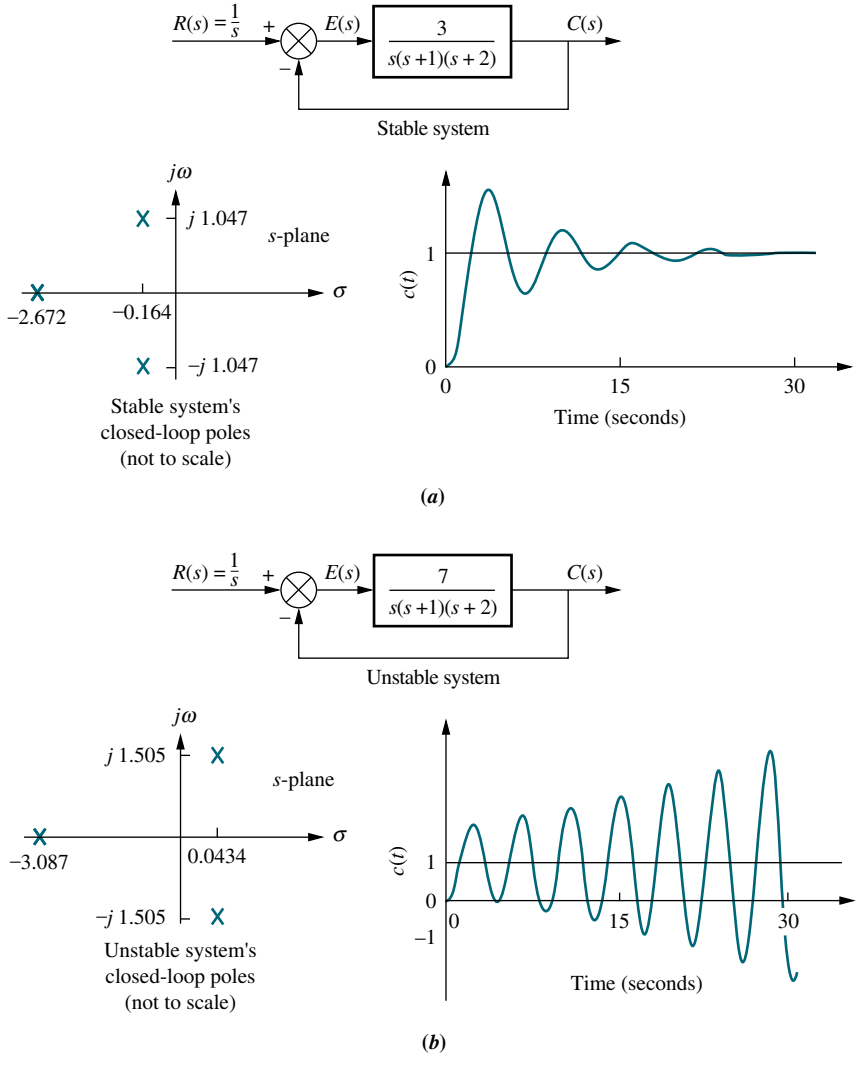


FIGURE 6.1 Closed-loop poles and response: **a.** stable system; **b.** unstable system

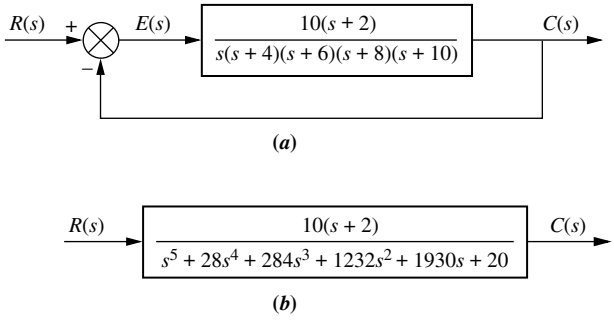


FIGURE 6.2 Common cause of problems in finding closed-loop poles: **a.** original system; **b.** equivalent system

left–half-plane poles, then the factors of the denominator of the closed-loop system transfer function consist of products of terms such as $(s + a_i)$, where a_i is real and positive, or complex with a positive real part. The product of such terms is a polynomial with all positive coefficients.[2] No term of the polynomial can be missing, since that would imply cancellation between positive and negative coefficients or imaginary axis roots in the factors, which is not the case. Thus, a sufficient condition for a system to be unstable is that all signs of the coefficients of the denominator of the closed-loop transfer function are not the same. If powers of s are missing, the system is either unstable or, at best, marginally stable. Unfortunately, if all coefficients of the denominator are positive and not missing, we do not have definitive information about the system's pole locations.

If the method described in the previous paragraph is not sufficient, then a computer can be used to determine the stability by calculating the root locations of the denominator of the closed-loop transfer function. Today some hand-held calculators can evaluate the roots of a polynomial. There is, however, another method to test for stability without having to solve for the roots of the denominator. We discuss this method in the next section.

6.2 Routh-Hurwitz Criterion

In this section, we learn a method that yields stability information without the need to solve for the closed-loop system poles. Using this method, we can tell how many closed-loop system poles are in the left half-plane, in the right half-plane, and on the $j\omega$-axis. (Notice that we say *how many*, not *where*.) We can find the number of poles in each section of the s-plane, but we cannot find their coordinates. The method is called the *Routh-Hurwitz criterion* for stability (*Routh*, *1905*).

The method requires two steps: (1) Generate a data table called a *Routh table* and (2) interpret the Routh table to tell how many closed-loop system poles are in the left half-plane, the right half-plane, and on the $j\omega$-axis. You might wonder why we study the Routh-Hurwitz criterion when modern calculators and computers can tell us the exact location of system poles. The power of the method lies in design rather than analysis. For example, if you have an unknown parameter in the denominator of a transfer function, it is difficult to determine via a calculator the range of this parameter to yield stability. You would probably rely on trial and error to answer the

[2] The coefficients can also be made all negative by multiplying the polynomial by −1. This operation does not change the root location.

stability question. We shall see later that the Routh-Hurwitz criterion can yield a closed-form expression for the range of the unknown parameter.

In this section, we make and interpret a basic Routh table. In the next section, we consider two special cases that can arise when generating this data table.

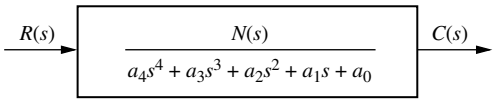


FIGURE 6.3 Equivalent closed-loop transfer function

Generating a Basic Routh Table

Look at the equivalent closed-loop transfer function shown in Figure 6.3. Since we are interested in the system poles, we focus our attention on the denominator. We first create the Routh table shown in Table 6.1. Begin by labeling the rows with powers of s from the highest power of the denominator of the closed-loop transfer function to s^0. Next start with the coefficient of the highest power of s in the denominator and list, horizontally in the first row, every other coefficient. In the second row, list horizontally, starting with the next highest power of s, every coefficient that was skipped in the first row.

The remaining entries are filled in as follows. Each entry is a negative determinant of entries in the previous two rows divided by the entry in the first column directly above the calculated row. The left-hand column of the determinant is always the first column of the previous two rows, and the right-hand column is the elements of the column above and to the right. The table is complete when all of the rows are completed down to s^0. Table 6.2 is the completed Routh table. Let us look at an example.

TABLE 6.1 Initial layout for Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2			
s^1			
s^0			

TABLE 6.2 Completed Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$-\dfrac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$-\dfrac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$-\dfrac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$-\dfrac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$-\dfrac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$-\dfrac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$-\dfrac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$-\dfrac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$-\dfrac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

Example 6.1

Creating a Routh Table

PROBLEM: Make the Routh table for the system shown in Figure 6.4(a).

SOLUTION: The first step is to find the equivalent closed-loop system because we want to test the denominator of this function, not the given forward transfer

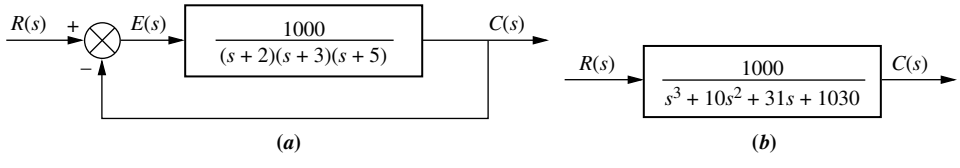


FIGURE 6.4 **a.** Feedback system for Example 6.1; **b.** equivalent closed-loop system

TABLE 6.3 Completed Routh table for Example 6.1

s^3	1	31	0
s^2	$\cancel{10}\ \ 1$	$\cancel{1030}\ \ 103$	0
s^1	$\dfrac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$
s^0	$\dfrac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$

function, for pole location. Using the feedback formula, we obtain the equivalent system of Figure 6.4(b). The Routh-Hurwitz criterion will be applied to this denominator. First label the rows with powers of s from s^3 down to s^0 in a vertical column, as shown in Table 6.3. Next form the first row of the table, using the coefficients of the denominator of the closed-loop transfer function. Start with the coefficient of the highest power and skip every other power of s. Now form the second row with the coefficients of the denominator skipped in the previous step. Subsequent rows are formed with determinants, as shown in Table 6.2.

For convenience, any row of the Routh table can be multiplied by a positive constant without changing the values of the rows below. This can be proved by examining the expressions for the entries and verifying that any multiplicative constant from a previous row cancels out. In the second row of Table 6.3, for example, the row was multiplied by 1/10. We see later that care must be taken not to multiply the row by a negative constant.

Interpreting the Basic Routh Table

Now that we know how to generate the Routh table, let us see how to interpret it. The basic Routh table applies to systems with poles in the left and right half-planes. Systems with imaginary poles and the kind of Routh table that results will be discussed in the next section. Simply stated, the Routh-Hurwitz criterion declares that *the number of roots of the polynomial that are in the right half-plane is equal to the number of sign changes in the first column*.

If the closed-loop transfer function has all poles in the left half of the s-plane, the system is stable. Thus, a system is stable if there are no sign changes in the first column of the Routh table. For example, Table 6.3 has two sign changes in the first column. The first sign change occurs from 1 in the s^2 row to -72 in the s^1 row. The second occurs from -72 in the s^1 row to 103 in the s^0 row. Thus, the system of Figure 6.4 is unstable since two poles exist in the right half-plane.

Skill-Assessment Exercise 6.1

PROBLEM: Make a Routh table and tell how many roots of the following polynomial are in the right half-plane and in the left half-plane.

$$P(s) = 3s^7 + 9s^6 + 6s^5 + 4s^4 + 7s^3 + 8s^2 + 2s + 6$$

ANSWER: Four in the right half-plane (rhp), three in the left half-plane (lhp).

The complete solution is at www.wiley.com/college/nise.

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Control Solutions

Now that we have described how to generate and interpret a basic Routh table, let us look at two special cases that can arise.

6.3 Routh-Hurwitz Criterion: Special Cases

Two special cases can occur: (1) The Routh table sometimes will have a zero *only in the first column* of a row, or (2) the Routh table sometimes will have an *entire row* that consists of zeros. Let us examine the first case.

Zero Only in the First Column

If the first element of a row is zero, division by zero would be required to form the next row. To avoid this phenomenon, an epsilon, ϵ, is assigned to replace the zero in the first column. The value ϵ is then allowed to approach zero from either the positive or the negative side, after which the signs of the entries in the first column can be determined. Let us look at an example.

Example 6.2

Stability via Epsilon Method

TryIt 6.1

Use the following MATLAB statement to find the poles of the closed-loop transfer function in Eq. (6.2).

```
roots([1 2 3 6 5 3])
```

PROBLEM: Determine the stability of the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3} \tag{6.2}$$

SOLUTION: The solution is shown in Table 6.4. We form the Routh table by using the denominator of Eq. (6.2). Begin by assembling the Routh table down to the row where a zero appears *only* in the first column (the s^3 row). Next replace the zero by a small number, ϵ, and complete the table. To begin the interpretation, we must first assume a sign, positive or negative, for the quantity ϵ. Table 6.5 shows the first column of Table 6.4 along with the resulting signs for choices of ϵ positive and ϵ negative.

TABLE 6.4 Completed Routh table for Example 6.2

s^5	1	3	5
s^4	2	6	3
s^3	ϵ	$\frac{7}{2}$	0
s^2	$\frac{6\epsilon - 7}{\epsilon}$	3	0
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	0	0
s^0	3	0	0

TABLE 6.5 Determining signs in first column of a Routh table with zero as first element in a row

Label	First column	$\epsilon = +$	$\epsilon = -$
s^5	1	+	+
s^4	2	+	+
s^3	ϵ	+	−
s^2	$\frac{6\epsilon - 7}{\epsilon}$	−	+
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	+	+
s^0	3	+	+

If ϵ is chosen positive, Table 6.5 will show a sign change from the s^3 row to the s^2 row, and there will be another sign change from the s^2 row to the s^1 row. Hence, the system is unstable and has two poles in the right half-plane.

Alternatively, we could choose ϵ negative. Table 6.5 would then show a sign change from the s^4 row to the s^3 row. Another sign change would occur from the s^3 row to the s^2 row. Our result would be exactly the same as that for a positive choice for ϵ. Thus, the system is unstable, with two poles in the right half-plane.

Symbolic Math SM

Students who are performing the MATLAB exercises and want to explore the added capability of MATLAB's Symbolic Math Toolbox should now run ch6sp1 in Appendix F at www.wiley.com/college/nise. You will learn how to use the Symbolic Math Toolbox to calculate the values of cells in a Routh table even if the table contains symbolic objects, such as ϵ. You will see that the Symbolic Math Toolbox and MATLAB yield an alternate way to generate the Routh table for Example 6.2.

Another method that can be used when a zero appears only in the first column of a row is derived from the fact that a polynomial that has the reciprocal roots of the original polynomial has its roots distributed the same—right half-plane, left half-plane, or imaginary axis—because taking the reciprocal of the root value does not move it to another region. Thus, if we can find the polynomial that has the reciprocal roots of the original, it is possible that the Routh table for the new polynomial will not have a zero in the first column. This method is usually computationally easier than the epsilon method just described.

We now show that the polynomial we are looking for, the one with the reciprocal roots, is simply the original polynomial with its coefficients written in reverse order (*Phillips, 1991*). Assume the equation

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0 \tag{6.3}$$

If s is replaced by $1/d$, then d will have roots which are the reciprocal of s. Making this substitution in Eq. (6.3),

$$\left(\frac{1}{d}\right)^n + a_{n-1}\left(\frac{1}{d}\right)^{n-1} + \cdots + a_1\left(\frac{1}{d}\right) + a_0 = 0 \tag{6.4}$$

Factoring out $(1/d)^n$,

$$\left(\frac{1}{d}\right)^n \left[1 + a_{n-1}\left(\frac{1}{d}\right)^{-1} + \cdots + a_1\left(\frac{1}{d}\right)^{(1-n)} + a_0\left(\frac{1}{d}\right)^{-n}\right]$$

$$= \left(\frac{1}{d}\right)^n [1 + a_{n-1}d + \cdots + a_1 d^{(n-1)} + a_0 d^n] = 0 \tag{6.5}$$

Thus, the polynomial with reciprocal roots is a polynomial with the coefficients written in reverse order. Let us redo the previous example to show the computational advantage of this method.

Example 6.3

Stability via Reverse Coefficients

PROBLEM: Determine the stability of the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3} \tag{6.6}$$

SOLUTION: First write a polynomial that has the reciprocal roots of the denominator of Eq. (6.6). From our discussion, this polynomial is formed by writing the denominator of Eq. (6.6) in reverse order. Hence,

$$D(s) = 3s^5 + 5s^4 + 6s^3 + 3s^2 + 2s + 1 \tag{6.7}$$

We form the Routh table as shown in Table 6.6 using Eq. (6.7). Since there are two sign changes, the system is unstable and has two right-half-plane poles. This is the same as the result obtained in Example 6.2. Notice that Table 6.6 does not have a zero in the first column.

TABLE 6.6 Routh table for Example 6.3

s^5	3	6	2
s^4	5	3	1
s^3	4.2	1.4	
s^2	1.33	1	
s^1	−1.75		
s^0	1		

Entire Row is Zero

We now look at the second special case. Sometimes while making a Routh table, we find that an entire row consists of zeros because there is an even polynomial that is a factor of the original polynomial. This case must be handled differently from the case of a zero in only the first column of a row. Let us look at an example that demonstrates how to construct and interpret the Routh table when an entire row of zeros is present.

Example 6.4

Stability via Routh Table with Row of Zeros

PROBLEM: Determine the number of right-half-plane poles in the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56} \tag{6.8}$$

SOLUTION: Start by forming the Routh table for the denominator of Eq. (6.8) (see Table 6.7). At the second row we multiply through by 1/7 for convenience. We stop at the third row, since the entire row consists of zeros, and use the following

TABLE 6.7 Routh table for Example 6.4

s^5			1			6			8
s^4		~~7~~	1		~~42~~	6		~~56~~	8
s^3	~~0~~	~~4~~	1	~~0~~	~~12~~	3	~~0~~	~~0~~	0
s^2			3			8			0
s^1			$\frac{1}{3}$			0			0
s^0			8			0			0

procedure. First we return to the row immediately above the row of zeros and form an auxiliary polynomial, using the entries in that row as coefficients. The polynomial will start with the power of s in the label column and continue by skipping every other power of s. Thus, the polynomial formed for this example is

$$P(s) = s^4 + 6s^2 + 8 \tag{6.9}$$

Next we differentiate the polynomial with respect to s and obtain

$$\frac{dP(s)}{ds} = 4s^3 + 12s + 0 \tag{6.10}$$

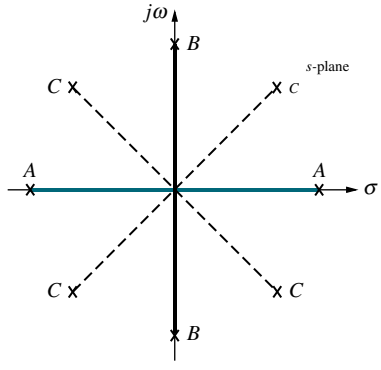
Finally, we use the coefficients of Eq. (6.10) to replace the row of zeros. Again, for convenience, the third row is multiplied by 1/4 after replacing the zeros.

The remainder of the table is formed in a straightforward manner by following the standard form shown in Table 6.2. Table 6.7 shows that all entries in the first column are positive. Hence, there are no right–half-plane poles.

Let us look further into the case that yields an entire row of zeros. An entire row of zeros will appear in the Routh table when a purely even or purely odd polynomial is a factor of the original polynomial. For example, $s^4 + 5s^2 + 7$ is an even polynomial; it has only even powers of s. Even polynomials only have roots that are symmetrical about the origin.[3] This symmetry can occur under three conditions of root position: (1) The roots are symmetrical and real, (2) the roots are symmetrical and imaginary, or (3) the roots are quadrantal. Figure 6.5 shows examples of these cases. Each case or combination of these cases will generate an even polynomial.

It is this even polynomial that causes the row of zeros to appear. Thus, the row of zeros tells us of the existence of an even polynomial whose roots are symmetric about the origin. Some of these roots could be on the $j\omega$-axis. On the other hand, since $j\omega$ roots are symmetric about the origin, if we do not have a row of zeros, we cannot possibly have $j\omega$ roots.

Another characteristic of the Routh table for the case in question is that the row previous to the row of zeros contains the even polynomial that is a factor of the original polynomial. Finally, everything from the row containing the even polynomial down to the end of the Routh table is a test of only the even polynomial. Let us put these facts together in an example.



A: Real and symmetrical about the origin
B: Imaginary and symmetrical about the origin
C: Quadrantal and symmetrical about the origin

FIGURE 6.5 Root positions to generate even polynomials: *A*, *B*, *C*, or any combination

[3] The polynomial $s^5 + 5s^3 + 7s$ is an example of an odd polynomial; it has only odd powers of s. Odd polynomials are the product of an even polynomial and an odd power of s. Thus, the constant term of an odd polynomial is always missing.

Example 6.5

Pole Distribution via Routh Table with Row of Zeros

PROBLEM: For the transfer function

$$T(s) = \frac{20}{s^8 + s^7 + 12s^6 + 22s^5 + 39s^4 + 59s^3 + 48s^2 + 38s + 20} \tag{6.11}$$

tell how many poles are in the right half-plane, in the left half-plane, and on the $j\omega$-axis.

SOLUTION: Use the denominator of Eq. (6.11) and form the Routh table in Table 6.8. For convenience the s^6 row is multiplied by 1/10, and the s^5 row is multiplied by 1/20. At the s^3 row we obtain a row of zeros. Moving back one row to s^4, we extract the even polynomial, $P(s)$, as

$$P(s) = s^4 + 3s^2 + 2 \tag{6.12}$$

TABLE 6.8 Routh table for Example 6.5

s^8	1	12	39	48	20
s^7	1	22	59	38	0
s^6	$-\cancel{10} - 1$	$-\cancel{20} - 2$	$\cancel{10}$ 1	$\cancel{20}$ 2	0
s^5	$\cancel{20}$ 1	$\cancel{60}$ 3	$\cancel{40}$ 2	0	0
s^4	1	3	2	0	0
s^3	$\cancel{0}$ $\cancel{4}$ 2	$\cancel{0}$ $\cancel{6}$ 3	$\cancel{0}$ $\cancel{0}$ 0	0	0
s^2	$\frac{3}{\cancel{2}}$ 3	$\cancel{2}$ 4	0	0	0
s^1	$\frac{1}{3}$	0	0	0	0
s^0	4	0	0	0	0

This polynomial will divide evenly into the denominator of Eq. (6.11) and thus is a factor. Taking the derivative with respect to s to obtain the coefficients that replace the row of zeros in the s^3 row, we find

$$\frac{dP(s)}{ds} = 4s^3 + 6s + 0 \tag{6.13}$$

Replace the row of zeros with 4, 6, and 0 and multiply the row by 1/2 for convenience. Finally, continue the table to the s^0 row, using the standard procedure.

How do we now interpret this Routh table? Since all entries from the even polynomial at the s^4 row down to the s^0 row are a test of the even polynomial, we begin to draw some conclusions about the roots of the even polynomial. No sign changes exist from the s^4 row down to the s^0 row. Thus, the even polynomial does not have right–half-plane poles. Since there are no right–half-plane poles, no left–half-plane poles are present because of the requirement for symmetry. Hence, the even polynomial, Eq. (6.12), must have all four of its poles on the $j\omega$-axis.[4] These results are summarized in the first column of Table 6.9.

[4] A necessary condition for stability is that the $j\omega$ roots have unit multiplicity. The even polynomial must be checked for multiple $j\omega$ roots. For this case, the existence of multiple $j\omega$ roots would lead to a perfect, fourth-order square polynomial. Since Eq. (6.12) is not a perfect square, the four $j\omega$ roots are distinct.

TABLE 6.9 Summary of pole locations for Example 6.5

	Polynomial		
Location	**Even (fourth-order)**	**Other (fourth-order)**	**Total (eighth-order)**
Right half-plane	0	2	2
Left half-plane	0	2	2
$j\omega$	4	0	4

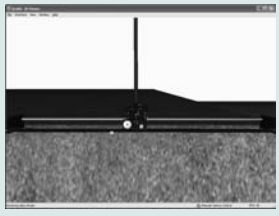
The remaining roots of the total polynomial are evaluated from the s^8 row down to the s^4 row. We notice two sign changes: one from the s^7 row to the s^6 row and the other from the s^6 row to the s^5 row. Thus, the other polynomial must have two roots in the right half-plane. These results are included in Table 6.9 under "Other". The final tally is the sum of roots from each component, the even polynomial and the other polynomial, as shown under "Total" in Table 6.9. Thus, the system has two poles in the right half-plane, two poles in the left half-plane, and four poles on the $j\omega$-axis; it is unstable because of the right–half-plane poles.

We now summarize what we have learned about polynomials that generate entire rows of zeros in the Routh table. These polynomials have a purely even factor with roots that are symmetrical about the origin. The even polynomial appears in the Routh table in the row directly above the row of zeros. Every entry in the table from the even polynomial's row to the end of the chart applies only to the even polynomial. Therefore, the number of sign changes from the even polynomial to the end of the table equals the number of right-half-plane roots of the even polynomial. Because of the symmetry of roots about the origin, the even polynomial must have the same number of left–half-plane roots as it does right–half-plane roots. Having accounted for the roots in the right and left half-planes, we know the remaining roots must be on the $j\omega$-axis.

Every row in the Routh table from the beginning of the chart to the row containing the even polynomial applies only to the other factor of the original polynomial. For this factor, the number of sign changes, from the beginning of the table down to the even polynomial, equals the number of right–half-plane roots. The remaining roots are left–half-plane roots. There can be no $j\omega$ roots contained in the other polynomial.

Virtual Experiment 6.1
Stability

Put theory into practice and evaluate the stability of the Quanser Linear Inverted Pendulum in LabVIEW. When in the upward balanced position, this system addresses the challenge of stabilizing a rocket during take-off. In the downward position it emulates the construction gantry crane.



Virtual experiments are found on WileyPLUS.

Skill-Assessment Exercise 6.2

PROBLEM: Use the Routh-Hurwitz criterion to find how many poles of the following closed-loop system, $T(s)$, are in the rhp, in the lhp, and on the $j\omega$-axis:

$$T(s) = \frac{s^3 + 7s^2 - 21s + 10}{s^6 + s^5 - 6s^4 + 0s^3 - s^2 - s + 6}$$

ANSWER: Two rhp, two lhp, and two $j\omega$

The complete solution is at www.wiley.com/college/nise.

Let us demonstrate the usefulness of the Routh-Hurwitz criterion with a few additional examples.

6.4 Routh-Hurwitz Criterion: Additional Examples

The previous two sections have introduced the Routh-Hurwitz criterion. Now we need to demonstrate the method's application to a number of analysis and design problems.

Example 6.6

Standard Routh-Hurwitz

PROBLEM: Find the number of poles in the left half-plane, the right half-plane, and on the $j\omega$-axis for the system of Figure 6.6.

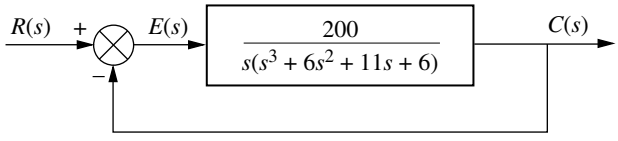


FIGURE 6.6 Feedback control system for Example 6.6

SOLUTION: First, find the closed-loop transfer function as

$$T(s) = \frac{200}{s^4 + 6s^3 + 11s^2 + 6s + 200} \tag{6.14}$$

The Routh table for the denominator of Eq. (6.14) is shown as Table 6.10. For clarity, we leave most zero cells blank. At the s^1 row there is a negative coefficient; thus, there are two sign changes. The system is unstable, since it has two right–half-plane poles and two left–half-plane poles. The system cannot have $j\omega$ poles since a row of zeros did not appear in the Routh table.

TABLE 6.10 Routh table for Example 6.6

s^4		1	11	200
s^3	~~6~~	1	~~6~~ 1	
s^2	~~10~~	1	~~200~~ 20	
s^1		−19		
s^0		20		

The next example demonstrates the occurrence of a zero in only the first column of a row.

Example 6.7

Routh-Hurwitz with Zero in First Column

PROBLEM: Find the number of poles in the left half-plane, the right half-plane, and on the $j\omega$-axis for the system of Figure 6.7.

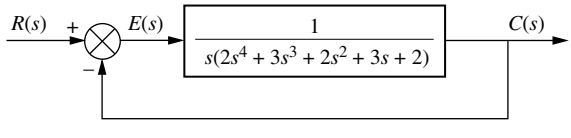


FIGURE 6.7 Feedback control system for Example 6.7

SOLUTION: The closed-loop transfer function is

$$T(s) = \frac{1}{2s^5 + 3s^4 + 2s^3 + 3s^2 + 2s + 1} \tag{6.15}$$

Form the Routh table shown as Table 6.11, using the denominator of Eq. (6.15). A zero appears in the first column of the s^3 row. Since the entire row is not zero, simply replace the zero with a small quantity, ϵ, and continue the table. Permitting ϵ to be a small, positive quantity, we find that the first term of the s^2 row is negative. Thus, there are two sign changes, and the system is unstable, with two poles in the right half-plane. The remaining poles are in the left half-plane.

TABLE 6.11 Routh table for Example 6.7

s^5	2	2	2
s^4	3	3	1
s^3	~~0~~ ϵ	$\frac{4}{3}$	
s^2	$\frac{3\epsilon - 4}{\epsilon}$	1	
s^1	$\frac{12\epsilon - 16 - 3\epsilon^2}{9\epsilon - 12}$		
s^0	1		

We also can use the alternative approach, where we produce a polynomial whose roots are the reciprocal of the original. Using the denominator of Eq. (6.15), we form a polynomial by writing the coefficients in reverse order,

$$s^5 + 2s^4 + 3s^3 + 2s^2 + 3s + 2 \tag{6.16}$$

The Routh table for this polynomial is shown as Table 6.12. Unfortunately, in this case we also produce a zero only in the first column at the s^2 row. However, the table is easier to work with than Table 6.11. Table 6.12 yields the same results as Table 6.11: three poles in the left half-plane and two poles in the right half-plane. The system is unstable.

TABLE 6.12 Alternative Routh table for Example 6.7

s^5	1	3	3
s^4	2	2	2
s^3	2	2	
s^2	~~0~~ ϵ	2	
s^1	$\frac{2\epsilon - 4}{\epsilon}$		
s^0	2		

MATLAB
ML

Students who are using MATLAB should now run ch6p1 in Appendix B. You will learn how to perform block diagram reduction to find $T(s)$, followed by an evaluation of the closed-loop system's poles to determine stability. This exercise uses MATLAB to do Example 6.7.

In the next example, we see an entire row of zeros appear along with the possibility of imaginary roots.

Example 6.8

Routh-Hurwitz with Row of Zeros

PROBLEM: Find the number of poles in the left half-plane, the right half-plane, and on the $j\omega$-axis for the system of Figure 6.8. Draw conclusions about the stability of the closed-loop system.

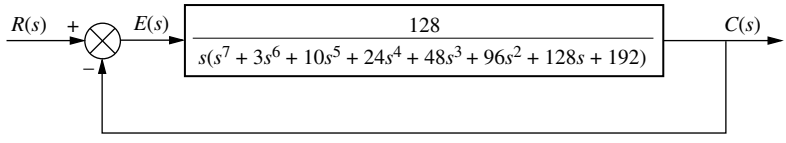


FIGURE 6.8 Feedback control system for Example 6.8

SOLUTION: The closed-loop transfer function for the system of Figure 6.8 is

$$T(s) = \frac{128}{s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128} \tag{6.17}$$

Using the denominator, form the Routh table shown as Table 6.13. A row of zeros appears in the s^5 row. Thus, the closed-loop transfer function denominator must have an even polynomial as a factor. Return to the s^6 row and form the even polynomial:

$$P(s) = s^6 + 8s^4 + 32s^2 + 64 \tag{6.18}$$

TABLE 6.13 Routh table for Example 6.8

s^8	1	10	48	128	128
s^7	~~3~~ 1	~~24~~ 8	~~96~~ 32	~~192~~ 64	
s^6	~~2~~ 1	~~16~~ 8	~~64~~ 32	~~128~~ 64	
s^5	~~0~~ ~~6~~ 3	~~0~~ ~~32~~ 16	~~0~~ ~~64~~ 32	~~0~~ ~~0~~ 0	
s^4	$\frac{8}{3}$ 1	$\frac{64}{3}$ 8	~~64~~ 24		
s^3	~~8~~ − 1	~~40~~ − 5			
s^2	~~3~~ 1	~~24~~ 8			
s^1	3				
s^0	8				

Differentiate this polynomial with respect to s to form the coefficients that will replace the row of zeros:

$$\frac{dP(s)}{ds} = 6s^5 + 32s^3 + 64s + 0 \tag{6.19}$$

Replace the row of zeros at the s^5 row by the coefficients of Eq. (6.19) and multiply through by 1/2 for convenience. Then complete the table.

We note that there are two sign changes from the even polynomial at the s^6 row down to the end of the table. Hence, the even polynomial has two right–half-

TryIt 6.2

Use MATLAB, The Control System Toolbox, and the following statements to find the closed-loop transfer function, $T(s)$, for Figure 6.8 and the closed-loop poles.

```
numg=128;
deng=[1 3 10 24 ...
 48 96 128 192 0];
G=tf(numg,deng);
T=feedback(G,1)
poles=pole(T)
```

TABLE 6.14 Summary of pole locations for Example 6.8

	Polynomial		
Location	**Even (sixth-order)**	**Other (second-order)**	**Total (eighth-order)**
Right half-plane	2	0	2
Left half-plane	2	2	4
$j\omega$	2	0	2

plane poles. Because of the symmetry about the origin, the even polynomial must have an equal number of left–half-plane poles. Therefore, the even polynomial has two left–half-plane poles. Since the even polynomial is of sixth order, the two remaining poles must be on the $j\omega$-axis.

There are no sign changes from the beginning of the table down to the even polynomial at the s^6 row. Therefore, the rest of the polynomial has no right–half-plane poles. The results are summarized in Table 6.14. The system has two poles in the right half-plane, four poles in the left half-plane, and two poles on the $j\omega$-axis, which are of unit multiplicity. The closed-loop system is unstable because of the right–half-plane poles.

The Routh-Hurwitz criterion gives vivid proof that changes in the gain of a feedback control system result in differences in transient response because of changes in closed-loop pole locations. The next example demonstrates this concept. We will see that for control systems, such as those shown in Figure 6.9, gain variations can move poles from stable regions of the s-plane onto the $j\omega$-axis and then into the right half-plane.

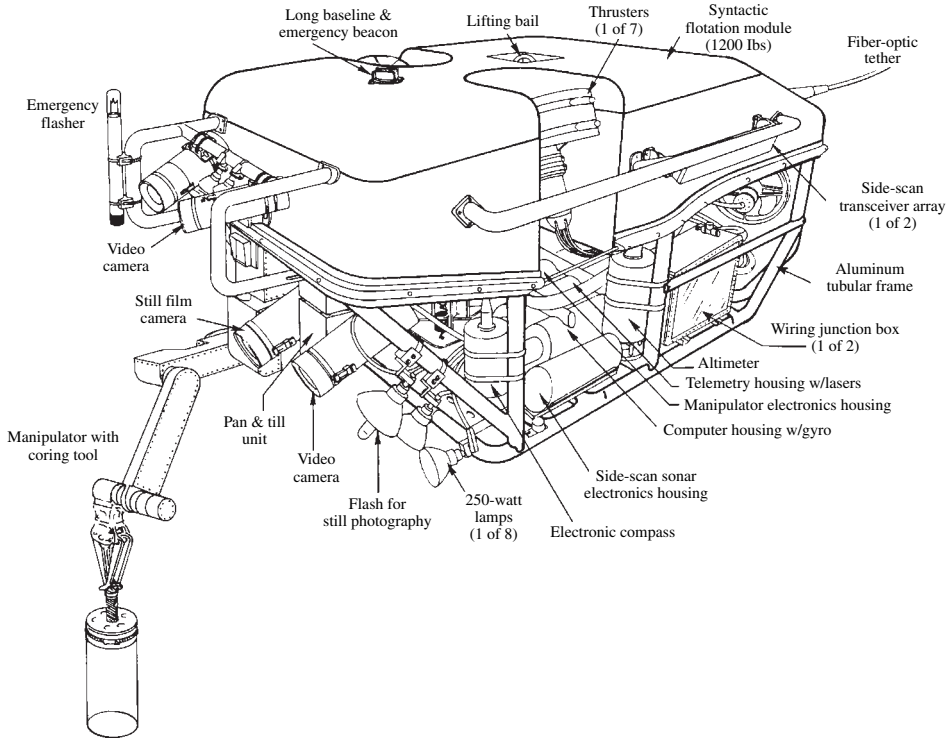


FIGURE 6.9 *Jason* is an underwater, remote-controlled vehicle that has been used to explore the wreckage of the *Lusitania*. The manipulator and cameras comprise some of the vehicle's control systems

Example 6.9

Stability Design via Routh-Hurwitz

PROBLEM: Find the range of gain, K, for the system of Figure 6.10 that will cause the system to be stable, unstable, and marginally stable. Assume $K > 0$.

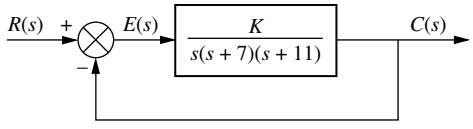


FIGURE 6.10 Feedback control system for Example 6.9

SOLUTION: First find the closed-loop transfer function as

$$T(s) = \frac{K}{s^3 + 18s^2 + 77s + K} \tag{6.20}$$

Next form the Routh table shown as Table 6.15.

TABLE 6.15 Routh table for Example 6.9

s^3	1	77
s^2	18	K
s^1	$\frac{1386 - K}{18}$	
s^0	K	

Since K is assumed positive, we see that all elements in the first column are always positive except the s^1 row. This entry can be positive, zero, or negative, depending upon the value of K. If $K < 1386$, all terms in the first column will be positive, and since there are no sign changes, the system will have three poles in the left half-plane and be *stable*.

If $K > 1386$, the s^1 term in the first column is negative. There are two sign changes, indicating that the system has two right–half-plane poles and one left–half-plane pole, which makes the system *unstable*.

If $K = 1386$, we have an entire row of zeros, which could signify $j\omega$ poles. Returning to the s^2 row and replacing K with 1386, we form the even polynomial

$$P(s) = 18s^2 + 1386 \tag{6.21}$$

Differentiating with respect to s, we have

$$\frac{dP(s)}{ds} = 36s + 0 \tag{6.22}$$

Replacing the row of zeros with the coefficients of Eq. (6.22), we obtain the Routh-Hurwitz table shown as Table 6.16 for the case of $K = 1386$.

TABLE 6.16 Routh table for Example 6.9 with $K = 1386$

s^3	1	77
s^2	18	1386
s^1	~~0~~ 36	
s^0	1386	

Since there are no sign changes from the even polynomial (s^2 row) down to the bottom of the table, the even polynomial has its two roots on the $j\omega$-axis of unit multiplicity. Since there are no sign changes above the even polynomial, the remaining root is in the left half-plane. Therefore the system is *marginally stable.*

MATLAB
ML

Students who are using MATLAB should now run ch6p2 in Appendix B. You will learn how to set up a loop to search for the range of gain to yield stability. This exercise uses MATLAB to do Example 6.9.

Symbolic Math
SM

Students who are performing the MATLAB exercises and want to explore the added capability of MATLAB's Symbolic Math Toolbox should now run ch6sp2 in Appendix F at www.wiley.com/college/nise. You will learn how to use the Symbolic Math Toolbox to calculate the values of cells in a Routh table even if the table contains symbolic objects, such as a variable gain, K. You will see that the Symbolic Math Toolbox and MATLAB yield an alternative way to solve Example 6.9.

The Routh-Hurwitz criterion is often used in limited applications to factor polynomials containing even factors. Let us look at an example.

Example 6.10

Factoring via Routh-Hurwitz

PROBLEM: Factor the polynomial

$$s^4 + 3s^3 + 30s^2 + 30s + 200 \tag{6.23}$$

SOLUTION: Form the Routh table of Table 6.17. We find that the s^1 row is a row of zeros. Now form the even polynomial at the s^2 row:

$$P(s) = s^2 + 10 \tag{6.24}$$

TABLE 6.17 Routh table for Example 6.10

s^4		1	30	200
s^3	~~3~~	1	~~30~~ 10	
s^2	~~20~~	1	~~200~~ 10	
s^1	~~0~~	2	~~0~~ 0	
s^0		10		

This polynomial is differentiated with respect to s in order to complete the Routh table. However, since this polynomial is a factor of the original polynomial in Eq. (6.23), dividing Eq. (6.23) by (6.24) yields $(s^2 + 3s + 20)$ as the other factor. Hence,

$$\begin{aligned} s^4 + 3s^3 + 30s^2 + 30s + 200 &= (s^2 + 10)(s^2 + 3s + 20) \\ &= (s + j3.1623)(s - j3.1623) \\ &\quad \times (s + 1.5 + j4.213)(s + 1.5 - j4.213) \end{aligned} \tag{6.25}$$

Skill-Assessment Exercise 6.3

WileyPLUS
WPCS
Control Solutions

PROBLEM: For a unity feedback system with the forward transfer function

$$G(s) = \frac{K(s+20)}{s(s+2)(s+3)}$$

find the range of K to make the system stable.

ANSWER: $0 < K < 2$

The complete solution is at www.wiley.com/college/nise.

6.5 Stability in State Space

State Space
SS

Up to this point we have examined stability from the s-plane viewpoint. Now we look at stability from the perspective of state space. In Section 4.10, we mentioned that the values of the system's poles are equal to the eigenvalues of the system matrix, $\mathbf{A}$. We stated that the eigenvalues of the matrix $\mathbf{A}$ were solutions of the equation $\det(s\mathbf{I} - \mathbf{A}) = 0$, which also yielded the poles of the transfer function. Eigenvalues appeared again in Section 5.8, where they were formally defined and used to diagonalize a matrix. Let us now formally show that the eigenvalues and the system poles have the same values.

Reviewing Section 5.8, the eigenvalues of a matrix, $\mathbf{A}$, are values of λ that permit a nontrivial solution (other than $\mathbf{0}$) for $\mathbf{x}$ in the equation

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \tag{6.26}$$

In order to solve for the values of λ that do indeed permit a solution for $\mathbf{x}$, we rearrange Eq. (6.26) as follows:

$$\lambda\mathbf{x} - \mathbf{A}\mathbf{x} = \mathbf{0} \tag{6.27}$$

or

$$(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0} \tag{6.28}$$

Solving for $\mathbf{x}$ yields

$$\mathbf{x} = (\lambda\mathbf{I} - \mathbf{A})^{-1}\mathbf{0} \tag{6.29}$$

or

$$\mathbf{x} = \frac{\text{adj}(\lambda\mathbf{I} - \mathbf{A})}{\det(\lambda\mathbf{I} - \mathbf{A})}\mathbf{0} \tag{6.30}$$

We see that all solutions will be the null vector except for the occurrence of zero in the denominator. Since this is the only condition where elements of $\mathbf{x}$ will be $0/0$, or indeterminate, it is the only case where a nonzero solution is possible.

The values of λ are calculated by forcing the denominator to zero:

$$\det(\lambda\mathbf{I} - \mathbf{A}) = 0 \tag{6.31}$$

This equation determines the values of λ for which a nonzero solution for $\mathbf{x}$ in Eq. (6.26) exists. In Section 5.8, we defined $\mathbf{x}$ as *eigenvectors* and the values of λ as the *eigenvalues* of the matrix $\mathbf{A}$.

Let us now relate the eigenvalues of the system matrix, $\mathbf{A}$, to the system's poles. In Chapter 3 we derived the equation of the system transfer function, Eq. (3.73), from the state equations. The system transfer function has $\det(s\mathbf{I} - \mathbf{A})$ in the denominator because of the presence of $(s\mathbf{I} - \mathbf{A})^{-1}$. Thus,

$$\det(s\mathbf{I} - \mathbf{A}) = 0 \tag{6.32}$$

is the characteristic equation for the system from which the system poles can be found.

Since Eqs. (6.31) and (6.32) are identical apart from a change in variable name, we conclude that the eigenvalues of the matrix $\mathbf{A}$ are identical to the system's poles before cancellation of common poles and zeroes in the transfer function. Thus, we can determine the stability of a system represented in state space by finding the eigenvalues of the system matrix, $\mathbf{A}$, and determining their locations on the s-plane.

Example 6.11

Stability in State Space

PROBLEM: Given the system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \tag{6.33a}$$

$$y = [1 \quad 0 \quad 0]\mathbf{x} \tag{6.33b}$$

find out how many poles are in the left half-plane, in the right half-plane, and on the $j\omega$-axis.

SOLUTION: First form $(s\mathbf{I} - \mathbf{A})$:

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} = \begin{bmatrix} s & -3 & -1 \\ -2 & s-8 & -1 \\ 10 & 5 & s+2 \end{bmatrix} \tag{6.34}$$

Now find the $\det(s\mathbf{I} - \mathbf{A})$:

$$\det(s\mathbf{I} - \mathbf{A}) = s^3 - 6s^2 - 7s - 52 \tag{6.35}$$

Using this polynomial, form the Routh table of Table 6.18.

TABLE 6.18 Routh table for Example 6.11

s^3	1	-7
s^2	~~-6~~ -3	~~-52~~ -26
s^1	~~$\frac{47}{3}$~~ -1	~~0~~ 0
s^0	-26	

Since there is one sign change in the first column, the system has one right–half-plane pole and two left–half-plane poles. It is therefore unstable. Yet, you may question the possibility that if a nonminimum-phase zero cancels the unstable pole, the system will be stable. However, in practice, the nonminimum-phase zero or unstable pole will shift due to a slight change in the system's parameters. This change will cause the system to become unstable.

MATLAB
ML

Students who are using MATLAB should now run ch6p3 in Appendix B. You will learn how to determine the stability of a system represented in state space by finding the eigenvalues of the system matrix. This exercise uses MATLAB to do Example 6.11.

Skill-Assessment Exercise 6.4

WileyPLUS
WPCS
Control Solutions

PROBLEM: For the following system represented in state space, find out how many poles are in the left half-plane, in the right half-plane, and on the $j\omega$-axis.

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 7 & 1 \\ -3 & 4 & -5 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$\mathrm{y} = [0 \quad 1 \quad 0]\mathbf{x}$$

ANSWER: Two rhp and one lhp.

The complete solution is at www.wiley.com/college/nise.

TryIt 6.3

Use the following MATLAB statements to find the eigenvalues of the system described in Skill-Assessment Exercise 6.4.

```
A=[2  1  1
     1  7  1
    −3  4  −5];
Eig=eig(A)
```

In this section, we have evaluated the stability of feedback control systems from the state-space perspective. Since the closed-loop poles and the eigenvalues of a system are the same, the stability requirement of a system represented in state space dictates that the eigenvalues cannot be in the right half of the s-plane or be multiple on the $j\omega$-axis.

We can obtain the eigenvalues from the state equations without first converting to a transfer function to find the poles: The equation $\det(s\mathbf{I} - \mathbf{A}) = 0$ yields the eigenvalues directly. If $\det(s\mathbf{I} - \mathbf{A})$, a polynomial in s, cannot be factored easily, we can apply the Routh-Hurwitz criterion to it to evaluate how many eigenvalues are in each region of the s-plane.

We now summarize this chapter, first with case studies and then with a written summary. Our case studies include the antenna azimuth position control system and the UFSS. Stability is as important to these systems as it is to the system shown in Figure 6.11.



FIGURE 6.11 The FANUC M-410iB™ has 4 axes of motion. It is seen here moving and stacking sacks of chocolate

Case Studies

Antenna Control: Stability Design via Gain

Design
D

This chapter has covered the elements of stability. We saw that stable systems have their closed-loop poles in the left half of the s-plane. As the loop gain is changed, the locations of the poles are also changed, creating the possibility that the poles can move into the right half of the s-plane, which yields instability. Proper gain settings are essential for the stability of closed-loop systems. The following case study demonstrates the proper setting of the loop gain to ensure stability.

PROBLEM: You are given the antenna azimuth position control system shown on the front endpapers, Configuration 1. Find the range of preamplifier gain required to keep the closed-loop system stable.

SOLUTION: The closed-loop transfer function was derived in the case studies in Chapter 5 as

$$T(s) = \frac{6.63K}{s^3 + 101.71s^2 + 171s + 6.63K} \tag{6.36}$$

Using the denominator, create the Routh table shown as Table 6.19. The third row of the table shows that a row of zeros occurs if $K = 2623$. This value of K makes the system marginally stable. Therefore, there will be no sign changes in the first column if $0 < K < 2623$. We conclude that, for stability, $0 < K < 2623$.

TABLE 6.19 Routh table for antenna control case study

s^3	1	171
s^2	101.71	$6.63K$
s^1	$17392.41–6.63K$	0
s^0	$6.63K$	

CHALLENGE: We now give you a problem to test your knowledge of this chapter's objectives. Refer to the antenna azimuth position control system shown on the front endpapers, Configuration 2. Find the range of preamplifier gain required to keep the closed-loop system stable.

UFSS Vehicle: Stability Design via Gain

Design
D

For this case study, we return to the UFSS vehicle and study the stability of the pitch control system, which is used to control depth. Specifically, we find the range of pitch gain that keeps the pitch control loop stable.

PROBLEM: The pitch control loop for the UFSS vehicle (*Johnson*, *1980*) is shown on the back endpapers. Let $K_2 = 1$ and find the range of K_1 that ensures that the closed-loop pitch control system is stable.

SOLUTION: The first step is to reduce the pitch control system to a single, closed-loop transfer function. The equivalent forward transfer function, $G_e(s)$, is

$$G_e(s) = \frac{0.25K_1(s + 0.435)}{s^4 + 3.456s^3 + 3.457s^2 + 0.719s + 0.0416} \tag{6.37}$$

With unity feedback the closed-loop transfer function, $T(s)$, is

$$T(s) = \frac{0.25K_1(s + 0.435)}{s^4 + 3.456s^3 + 3.457s^2 + (0.719 + 0.25K_1)s + (0.0416 + 0.109K_1)} \tag{6.38}$$

The denominator of Eq. (6.38) is now used to form the Routh table shown as Table 6.20.

TABLE 6.20 Routh table for UFSS case study

s^4	1	3.457	$0.0416 + 0.109K_1$
s^3	3.456	$0.719 + 0.25K_1$	
s^2	$11.228 - 0.25K_1$	$0.144 + 0.377K_1$	
s^1	$\dfrac{-0.0625K_1^2 + 1.324K_1 + 7.575}{11.228 - 0.25K_1}$		
s^0	$0.144 + 0.377K_1$		

Note: Some rows have been multiplied by a positive constant for convenience.

Looking at the first column, the s^4 and s^3 rows are positive. Thus, all elements of the first column must be positive for stability. For the first column of the s^2 row to be positive, $-\infty < K_1 < 44.91$. For the first column of the s^1 row to be positive, the numerator must be positive, since the denominator is positive from the previous step. The solution to the quadratic term in the numerator yields roots of $K_1 = -4.685$ and 25.87. Thus, for a positive numerator, $-4.685 < K_1 < 25.87$. Finally, for the first column of the s^0 row to be positive, $-0.382 < K_1 < \infty$. Using all three conditions, stability will be ensured if $-0.382 < K_1 < 25.87$.

CHALLENGE: You are now given a problem to test your knowledge of this chapter's objectives. For the UFSS vehicle (*Johnson*, *1980*) heading control system shown on the back endpapers and introduced in the UFSS case study challenge in Chapter 5, do the following:

MATLAB
ML

a. Find the range of heading gain that ensures the vehicle's stability. Let $K_2 = 1$

b. `Repeat Part` **a** `using MATLAB.`

In our case studies, we calculated the ranges of gain to ensure stability. The student should be aware that although these ranges yield stability, setting gain within these limits may not yield the desired transient response or steady-state error characteristics. In Chapters 9 and 11, we will explore design techniques, other than simple gain adjustment, that yield more flexibility in obtaining desired characteristics.

Summary

In this chapter, we explored the concepts of system stability from both the classical and the state-space viewpoints. We found that for linear systems, *stability* is based on a natural response that decays to zero as time approaches infinity. On the other hand, if the natural response increases without bound, the forced response is overpowered by the natural response, and we lose control. This condition is known as *instability*. A third possibility exists: The natural response may neither decay nor grow without bound but oscillate. In this case, the system is said to be *marginally stable*.

We also used an alternative definition of stability when the natural response is not explicitly available. This definition is based on the total response and says that a system is stable if every bounded input yields a bounded output (BIBO) and unstable if any bounded input yields an unbounded output.

Mathematically, stability for linear, time-invariant systems can be determined from the location of the closed-loop poles:

- If the poles are only in the left half-plane, the system is stable.
- If any poles are in the right half-plane, the system is unstable.
- If the poles are on the $j\omega$-axis and in the left half-plane, the system is marginally stable as long as the poles on the $j\omega$-axis are of unit multiplicity; it is unstable if there are any multiple $j\omega$ poles.

Unfortunately, although the open-loop poles may be known, we found that in higher-order systems it is difficult to find the closed-loop poles without a computer program.

The *Routh-Hurwitz criterion* lets us find how many poles are in each of the sections of the s-plane without giving us the coordinates of the poles. Just knowing that there are poles in the right half-plane is enough to determine that a system is unstable. Under certain limited conditions, when an even polynomial is present, the Routh table can be used to factor the system's characteristic equation.

Obtaining stability from the state-space representation of a system is based on the same concept—the location of the roots of the characteristic equation. These roots are equivalent to the eigenvalues of the system matrix and can be found by solving $\det(s\mathbf{I} - \mathbf{A}) = 0$. Again, the Routh-Hurwitz criterion can be applied to this polynomial. The point is that the state-space representation of a system need not be converted to a transfer function in order to investigate stability. In the next chapter, we will look at steady-state errors, the last of three important control system requirements we emphasize.

Review Questions

1. What part of the output response is responsible for determining the stability of a linear system?
2. What happens to the response named in Question 1 that creates instability?

3. What would happen to a physical system that becomes unstable?
4. Why are marginally stable systems considered unstable under the BIBO definition of stability?
5. Where do system poles have to be to ensure that a system is not unstable?
6. What does the Routh-Hurwitz criterion tell us?
7. Under what conditions would the Routh-Hurwitz criterion easily tell us the actual location of the system's closed-loop poles?
8. What causes a zero to show up only in the first column of the Routh table?
9. What causes an entire row of zeros to show up in the Routh table?
10. Why do we sometimes multiply a row of a Routh table by a positive constant?
11. Why do we not multiply a row of a Routh table by a negative constant?
12. If a Routh table has two sign changes above the even polynomial and five sign changes below the even polynomial, how many right–half-plane poles does the system have?
13. Does the presence of an entire row of zeros always mean that the system has $j\omega$ poles?
14. If a seventh-order system has a row of zeros at the s^3 row and two sign changes below the s^4 row, how many $j\omega$ poles does the system have?

State Space SS

15. Is it true that the eigenvalues of the system matrix are the same as the closed-loop poles?

State Space SS

16. How do we find the eigenvalues?

Problems

1. Tell how many roots of the following polynomial are in the right half-plane, in the left half-plane, and on the $j\omega$-axis: [Section: 6.2]

$$P(s) = s^5 + 3s^4 + 5s^3 + 4s^2 + s + 3$$

2. Tell how many roots of the following polynomial are in the right half-plane, in the left half-plane, and on the $j\omega$-axis: [Section: 6.3]

$$P(s) = s^5 + 6s^3 + 5s^2 + 8s + 20$$

3. Using the Routh table, tell how many poles of the following function are in the right half-plane, in the left half-plane, and on the $j\omega$-axis: [Section: 6.3]

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$$T(s) = \frac{s + 8}{s^5 - s^4 + 4s^3 - 4s^2 + 3s - 2}$$

4. The closed-loop transfer function of a system is [Section: 6.3]

$$T(s) = \frac{s^3 + 2s^2 + 7s + 21}{s^5 - 2s^4 + 3s^3 - 6s^2 + 2s - 4}$$

Determine how many closed-loop poles lie in the right half-plane, in the left half-plane, and on the $j\omega$-axis.

5. How many poles are in the right half-plane, in the left half-plane, and on the $j\omega$-axis for the open-loop system of Figure P6.1? [Section: 6.3]

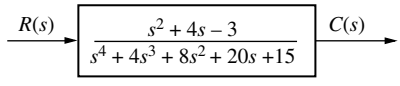


FIGURE P6.1

6. How many poles are in the right half-plane, the left half-plane, and on the $j\omega$-axis for the open-loop system of Figure P6.2? [Section: 6.3]

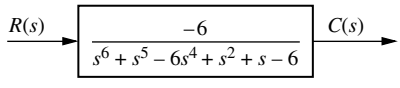


FIGURE P6.2

7. Use MATLAB to find the pole locations for the system of Problem 6.

MATLAB ML

8. Use MATLAB and the Symbolic Math Toolbox to generate a Routh table to solve Problem 3.

Symbolic Math SM

9. Determine whether the unity feedback system of Figure P6.3 is stable if [Section: 6.2]

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$$G(s) = \frac{240}{(s+1)(s+2)(s+3)(s+4)}$$

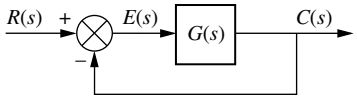


FIGURE P6.3

10. Use MATLAB to find the pole locations for the system of Problem 9.

MATLAB ML

11. Consider the unity feedback system of Figure P6.3 with

$$G(s) = \frac{1}{4s^2(s^2+1)}$$

Using the Routh-Hurwitz criterion, find the region of the s-plane where the poles of the closed-loop system are located. [Section: 6.3]

12. In the system of Figure P6.3, let

$$G(s) = \frac{K(s+2)}{s(s-1)(s+3)}$$

Find the range of K for closed-loop stability. [Section: 6.4]

13. Given the unity feedback system of Figure P6.3 with [Section: 6.3]

$$G(s) = \frac{84}{s(s^7+5s^6+12s^5+25s^4+45s^3+50s^2+82s+60)}$$

tell how many poles of the closed-loop transfer function lie in the right half-plane, in the left half-plane, and on the $j\omega$-axis. [Section: 6.3]

14. Using the Routh-Hurwitz criterion and the unity feedback system of Figure P6.3 with

$$G(s) = \frac{1}{2s^4+5s^3+s^2+2s}$$

tell whether or not the closed-loop system is stable. [Section: 6.2]

15. Given the unity feedback system of Figure P6.3 with

$$G(s) = \frac{8}{s(s^6-2s^5-s^4+2s^3+4s^2-8s-4)}$$

tell how many closed-loop poles are located in the right half-plane, in the left half-plane, and on the $j\omega$-axis. [Section: 6.3]

16. Repeat Problem 15 using MATLAB.

MATLAB ML

17. Consider the following Routh table. Notice that the s^5 row was originally all zeros. Tell how many roots of the original polynomial were in the right half-plane, in the left half-plane, and on the $j\omega$-axis. [Section: 6.3]

s^7	1	2	−1	−2
s^6	1	2	−1	−2
s^5	3	4	−1	0
s^4	1	−1	−3	0
s^3	7	8	0	0
s^2	−15	−21	0	0
s^1	−9	0	0	0
s^0	−21	0	0	0

18. For the system of Figure P6.4, tell how many closed-loop poles are located in the right half-plane, in the left half-plane, and on the $j\omega$-axis. Notice that there is positive feedback. [Section: 6.3]

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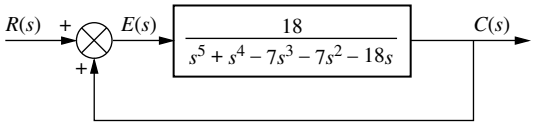


FIGURE P6.4

19. Using the Routh-Hurwitz criterion, tell how many closed-loop poles of the system shown in Figure P6.5 lie in the left half-plane, in the right half-plane, and on the $j\omega$-axis. [Section: 6.3]

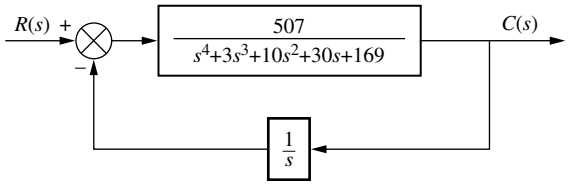


FIGURE P6.5

20. Determine if the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s^2+1)}{(s+1)(s+2)}$$

can be unstable. [Section: 6.4]

21. For the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s+6)}{s(s+1)(s+4)}$$

determine the range of K to ensure stability. [Section: 6.4]

22. In the system of Figure P6.3, let

$$G(s) = \frac{K(s-a)}{s(s-b)}$$

Find the range of K for closed-loop stability when: [Section: 6.4]

a. $a < 0, \quad b < 0$

b. $a < 0, \quad b > 0$

c. $a > 0, \quad b < 0$

d. $a > 0, \quad b > 0$

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23. For the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s+3)(s+5)}{(s-2)(s-4)}$$

determine the range of K for stability. [Section: 6.4]

MATLAB ML

24. Repeat Problem 23 using MATLAB.

Symbolic Math SM

25. Use MATLAB and the Symbolic Math Toolbox to generate a Routh table in terms of K to solve Problem 23.

26. Find the range of K for stability for the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K(s+4)(s-4)}{(s^2+3)}$$

27. For the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s+1)}{s^4(s+2)}$$

find the range of K for stability. [Section: 6.4]

28. Find the range of gain, K, to ensure stability in the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K(s-2)(s+4)(s+5)}{(s^2+12)}$$

29. Find the range of gain, K, to ensure stability in the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K(s+2)}{(s^2+1)(s+4)(s-1)}$$

30. Using the Routh-Hurwitz criterion, find the value of K that will yield oscillations for the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K}{(s+77)(s+27)(s+38)}$$

31. Use the Routh-Hurwitz criterion to find the range of K for which the system of Figure P6.6 is stable. [Section: 6.4]

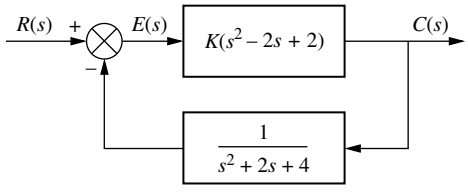


FIGURE P6.6

32. Repeat Problem 31 for the system of Figure P6.7. [Section: 6.4]

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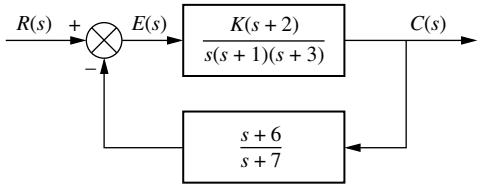


FIGURE P6.7

33. Given the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s+4)}{s(s+1.2)(s+2)}$$

find the following: [Section: 6.4]

a. The range of K that keeps the system stable

b. The value of K that makes the system oscillate

c. The frequency of oscillation when K is set to the value that makes the system oscillate

34. Repeat Problem 33 for [Section: 6.4]

$$G(s) = \frac{K(s-1)(s-2)}{(s+2)(s^2+2s+2)}$$

35. For the system shown in Figure P6.8, find the value of gain, K, that will make the system oscillate. Also, find the frequency of oscillation. [Section: 6.4]

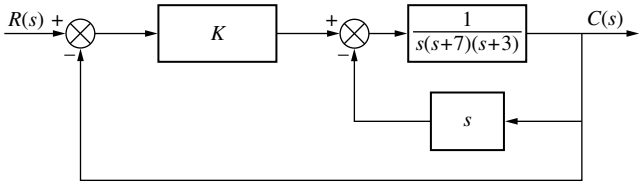


FIGURE P6.8

36. Given the unity feedback system of Figure P6.3 with [Section: 6.4]

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$$G(s) = \frac{Ks(s+2)}{(s^2-4s+8)(s+3)}$$

a. Find the range of K for stability.

b. Find the frequency of oscillation when the system is marginally stable.

37. Repeat Problem 36 using MATLAB.

MATLAB ML

38. For the unity feedback system of Figure P6.3 with

$$G(s) = \frac{K(s+2)}{(s^2+1)(s+4)(s-1)}$$

find the range of K for which there will be only two closed-loop, right–half-plane poles. [Section: 6.4]

39. For the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K}{(s+1)^3(s+4)}$$

a. Find the range of K for stability.

b. Find the frequency of oscillation when the system is marginally stable.

40. Given the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K}{(s+49)(s^2+4s+5)}$$

a. Find the range of K for stability.

b. Find the frequency of oscillation when the system is marginally stable.

41. Using the Routh-Hurwitz criterion and the unity feedback system of Figure P6.3 with [Section: 6.4]

$$G(s) = \frac{K}{s(s+1)(s+2)(s+5)}$$

a. Find the range of K for stability.

b. Find the value of K for marginal stability.

c. Find the actual location of the closed-loop poles when the system is marginally stable.

42. Find the range of K to keep the system shown in Figure P6.9 stable. [Section: 6.4]

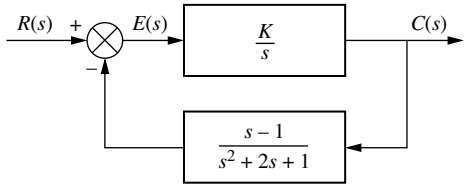


FIGURE P6.9

43. Find the value of K in the system of Figure P6.10 that will place the closed-loop poles as shown. [Section: 6.4]

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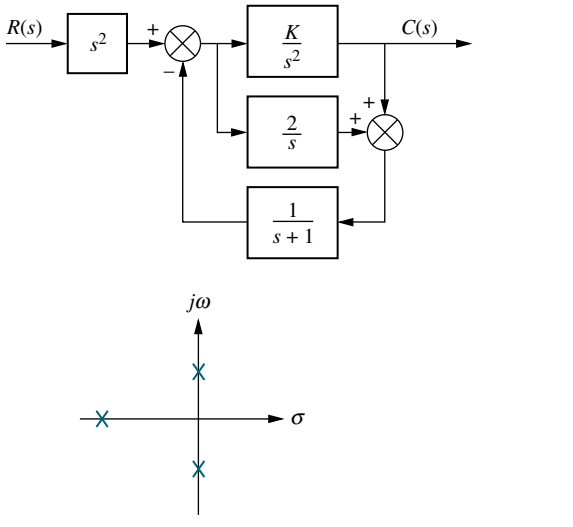


FIGURE P6.10 Closed-loop system with pole plot

44. The closed-loop transfer function of a system is

$$T(s) = \frac{s^2 + K_1 s + K_2}{s^4 + K_1 s^3 + K_2 s^2 + 5s + 1}$$

Determine the range of K_1 in order for the system to be stable. What is the relationship between K_1 and K_2 for stability? [Section: 6.4]

45. For the transfer function below, find the constraints on K_1 and K_2 such that the function will have only two $j\omega$ poles. [Section: 6.4]

$$T(s) = \frac{K_1 s + K_2}{s^4 + K_1 s^3 + s^2 + K_2 s + 1}$$

46. The transfer function relating the output engine fan speed (rpm) to the input main burner fuel flow rate (lb/h) in a short takeoff and landing (STOL) fighter aircraft, ignoring the coupling between engine fan speed and the pitch control command, is (*Schierman, 1992*) [Section: 6.4]

$$G(s) = \frac{1.3s^7 + 90.5s^6 + 1970s^5 + 15,000s^4 + 3120s^3 - 41,300s^2 - 5000s - 1840}{s^8 + 103s^7 + 1180s^6 + 4040s^5 + 2150s^4 - 8960s^3 - 10,600s^2 - 1550s - 415}$$

a. Find how many poles are in the right half-plane, in the left half-plane, and on the $j\omega$-axis.

b. Is this open-loop system stable?

47. An interval polynomial is of the form

$$P(s) = a_0 + a_1 s + a_2 s^2 + a_3 s^3 + a_4 s^4 + a_5 s^5 + \cdots$$

with its coefficients belonging to intervals $x_i \le a_i \le y_i$, where x_i, y_i are prescribed constants. Kharitonov's theorem says that an interval polynomial has all its roots in the left half-plane if each one of the following four polynomials has its roots in the left half-plane (*Minichelli, 1989*):

$$K_1(s) = x_0 + x_1 s + y_2 s^2 + y_3 s^3 + x_4 s^4 + x_5 s^5 + y_6 s^6 + \cdots$$
$$K_2(s) = x_0 + y_1 s + y_2 s^2 + x_3 s^3 + x_4 s^4 + y_5 s^5 + y_6 s^6 + \cdots$$
$$K_3(s) = y_0 + x_1 s + x_2 s^2 + y_3 s^3 + y_4 s^4 + x_5 s^5 + x_6 s^6 + \cdots$$
$$K_4(s) = y_0 + y_1 s + x_2 s^2 + x_3 s^3 + y_4 s^4 + y_5 s^5 + x_6 s^6 + \cdots$$

Use Kharitonov's theorem and the Routh-Hurwitz criterion to find if the following polynomial has any zeros in the right–half-plane.

$$P(s) = a_0 + a_1 s + a_2 s^2 + a_3 s^3$$
$$2 \le a_0 \le 4; \quad 1 \le a_1 \le 2; \quad 4 \le a_2 \le 6; \quad a_3 = 1$$

48. A linearized model of a torque-controlled crane hoisting a load with a fixed rope length is

$$P(s) = \frac{X_T(s)}{F_T(s)} = \frac{1}{m_T} \frac{s^2 + \omega_0^2}{s^2(s^2 + a\omega_0^2)}$$

where $\omega_0 = \sqrt{\frac{g}{L}}$, L = the rope length, m_T = the mass of the car, a = the combined rope and car mass, f_T = the force input applied to the car, and x_T = the resulting rope displacement (*Marttinen, 1990*). If the system is controlled in a feedback configuration by placing it in a loop as shown in Figure P6.11, with $K > 0$, where will the closed-loop poles be located?

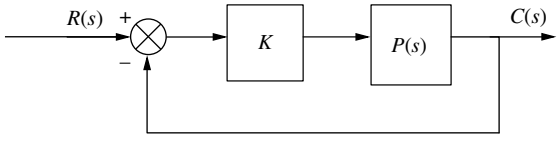


FIGURE P6.11

49. The read/write head assembly arm of a computer hard disk drive (HDD) can be modeled as a rigid rotating body with inertia I_b. Its dynamics can be described with the transfer function

$$P(s) = \frac{X(s)}{F(s)} = \frac{1}{I_b s^2}$$

where $X(s)$ is the displacement of the read/write head and $F(s)$ is the applied force (*Yan, 2003*). Show that if the HDD is controlled in the configuration shown in Figure P6.11, the arm will oscillate and cannot be positioned with any precision over a HDD track. Find the oscillation frequency.

50. A system is represented in state space as State Space SS

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 2 & -4 \\ 1 & -4 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$
$$y = [1 \quad 1 \quad 0]\mathbf{x}$$

Determine how many eigenvalues are in the right half-plane, in the left half-plane, and on the $j\omega$-axis. [Section: 6.5]

51. Use MATLAB to find the eigenvalues of the following system:

MATLAB ML State Space SS

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -4 \\ -1 & 1 & 8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = [0 \quad 0 \quad 1]\mathbf{x}$$

52. The following system in state space represents the forward path of a unity feedback system. Use the Routh-Hurwitz criterion to determine if the closed-loop system is stable. [Section: 6.5]

State Space SS WileyPLUS WPCS Control Solutions

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 3 \\ -3 & -4 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [0 \quad 1 \quad 1]\mathbf{x}$$

53. Repeat Problem 52 using MATLAB.

MATLAB ML

54. A Butterworth polynomial is of the form

$$B_n(s) = 1 + (-1)^n \left(\frac{s}{\omega_c}\right)^{2n}, \quad n > 0.$$

Use the Routh-Hurwitz criteria to find the zeros of a Butterworth polynomial for:

a. $n = 1$;

b. $n = 2$

DESIGN PROBLEMS

55. A model for an airplane's pitch loop is shown in Figure P6.12. Find the range of gain, K, that will keep the system stable. Can the system ever be unstable for positive values of K?

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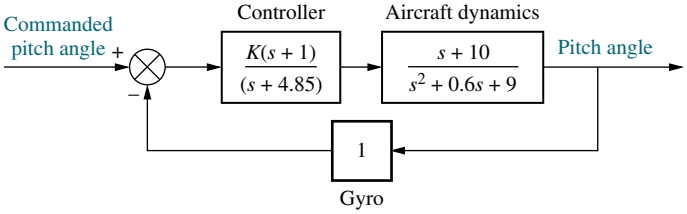


FIGURE P6.12 Aircraft pitch loop model

56. A common application of control systems is in regulating the temperature of a chemical process (Figure P6.13). The flow of a chemical reactant to a process is controlled by an actuator and valve. The reactant causes the temperature in the vat to change. This temperature is sensed and compared to a desired set-point temperature in a closed loop, where the flow of reactant is adjusted to yield the desired temperature. In Chapter 9, we will learn how a PID controller is used to improve the performance of such process control systems. Figure P6.13 shows the control system prior to the addition of the PID controller. The PID controller is replaced by the shaded box with a gain of unity. For this system, prior to the design of the PID controller, find the range of amplifier gain, K, to keep the system stable.

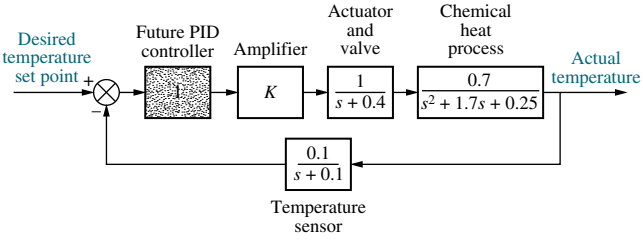
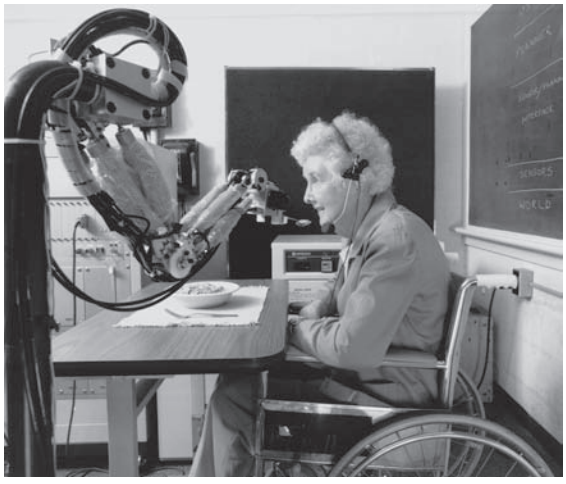
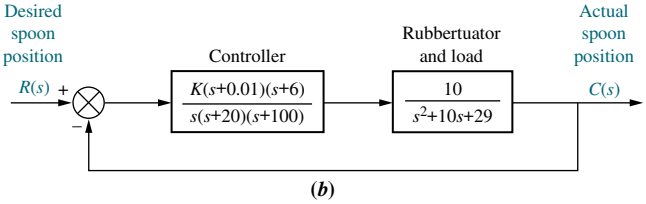


FIGURE P6.13 Block diagram of a chemical process control system

57. A robot arm called ISAC (Intelligent Soft Arm Control) can be used as part of a system to feed people with disabilities (see Figure P6.14(*a*)). The control system guides the spoon to the food and then to a position near the person's mouth. The arm uses a special pneumatically controlled actuator called a rubbertuator. Rubbertuators consist of rubber tubes covered with fiber cord. The actuator contracts in length when pneumatic pressure is increased and expands in length when pressure is decreased. This expansion and contraction in length can drive a pulley or other device. A video camera provides the sight for the robot and the tracking loop (*Kara, 1992*). Assume the simplified block diagram shown in Figure P6.14(*b*) for regulating the spoon at a distance from the mouth. Find the range of K for stability. (Use of a program with symbolic capability is recommended.)



(*a*)



(*b*)

FIGURE P6.14 **a.** ISAC used for feeding (Courtesy of Kazuhiko Kawamura, Vanderbilt University.) **b.** simplified block diagram

58. Often an aircraft is required to tow another vehicle, such as a practice target or glider. To stabilize the towed vehicle and prevent it from rolling, pitching, and yawing, an autopilot is built into the towed vehicle. Assume the block diagram shown in Figure P6.15 represents the autopilot roll control system (*Cochran*, *1992*). Find the range of K to keep the roll angle stable.

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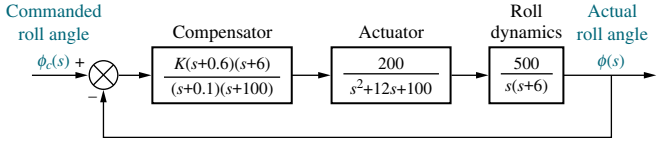


FIGURE P6.15 Towed vehicle roll control

59. Cutting forces should be kept constant during machining operations to prevent changes in spindle speeds or work position. Such changes would deteriorate the accuracy of the work's dimensions. A control system is proposed to control the cutting force. The plant is difficult to model, since the factors that affect cutting force are time varying and not easily predicted. However, assuming the simplified force control model shown in Figure P6.16, use the Routh-Hurwitz criterion to find the range of K to keep the system stable (*Rober*, *1997*).

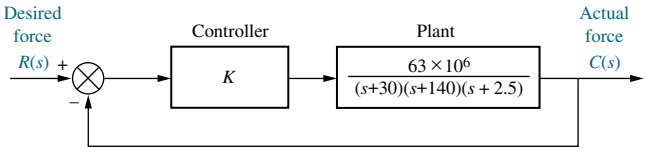
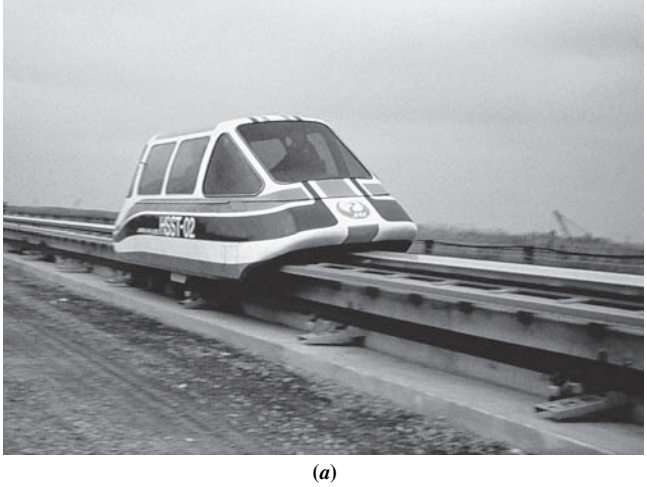
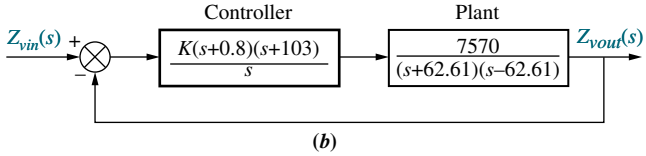


FIGURE P6.16 Cutting force control system (Reprinted with permission of ASME.)

60. Transportation systems that use magnetic levitation can reach very high speeds, since contact friction at the rails is eliminated (see Figure P6.17(*a*)). Electromagnets can produce the force to elevate the vehicle. Figure P6.17(*b*) is a simulation model of a control system that can be used to regulate the magnetic gap. In the figure, $Z_{vin}(s)$ represents a voltage proportional to the desired amount of levitation, or gap. $Z_{vout}(s)$ represents a voltage proportional to the actual amount of levitation. The plant models the dynamic response of the vehicle to



(*a*)



(*b*)

FIGURE P6.17 **a.** A magnetic levitation transportation system (© Japan Air Lines/Photo Researchers); **b.** simplified block diagram (© 1998 IEEE)

signals from the controller (*Bittar*, *1998*). Use the Routh-Hurwitz criterion to find the range of gain, K, to keep the closed loop system stable.

61. A transfer function from indoor radiator power, $\dot{Q}(s)$, to room temperature, $T(s)$, in an 11 m^2 room is

$$P(s) = \frac{T(s)}{\dot{Q}(s)} = \frac{1\times10^{-6}s^2 + 1.314\times10^{-9}s + 2.66\times10^{-13}}{s^3 + 0.00163s^2 + 5.272\times10^{-7}s + 3.538\times10^{-11}}$$

where $\dot{Q}$ is in watts and T is in °C. (*Thomas*, *2005*). The room's temperature will be controlled by embedding it in a closed loop, such as that of Figure P6.11. Find the range of K for closed-loop stability.

62. During vertical spindle surface grinding, adjustments are made on a multi-axis computer numerical control (CNC) machine by measuring the applied force with a dynamometer and applying appropriate corrections. This feedback force control results in higher homogeneity and better tolerances in the resulting finished product. In a specific experiment with an extremely high feed rate, the transfer function from the desired depth of cut (DOC) to applied force was

$$\frac{F(s)}{DOC(s)} = \frac{K_C}{1 + \dfrac{K_C}{ms^2 + bs + k} - \dfrac{K_C}{K_f}\dfrac{1}{Ts+1}}$$

where $k = 2.1\times10^4$ N/m, $b = 0.78$ Ns/m, $m = 1.2\times10^{-4}$ Kg, $K_C = 1.5\times10^4$ N/mm and $T = 0.004$ s. K_f is a parameter that is varied to adjust the system. Find the range of K_f under which the system is stable (*Hekman*, *1999*).

63. Figure P6.18 depicts the schematic diagram of a phase shift oscillator.

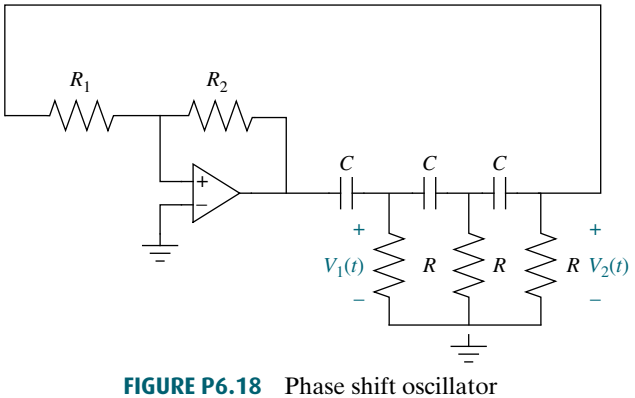


FIGURE P6.18 Phase shift oscillator

The circuit will oscillate if it is designed to have poles on the $j\omega$-axis.

a. Show that the transfer function for the passive network in the circuit is given by

$$\frac{V_2(s)}{V_1(s)} = \frac{-1}{\left(1+\dfrac{1}{sRC}\right)\left(2+\dfrac{1}{sRC}\right)^2 - 3 - \dfrac{2}{sRC}}$$

b. Show that the oscillator's characteristic equation is given by

$$1 - K\frac{1}{\left(1+\dfrac{1}{sRC}\right)\left(2+\dfrac{1}{sRC}\right)^2 - 3 - \dfrac{2}{sRC}} = 0,$$

where $K = \dfrac{R_2}{R_1}$

c. Use the Routh-Hurwitz criterion to obtain the oscillation condition and the oscillation frequency.

64. In order to obtain a low-cost lithium-ion battery charger, the feedback loop of Figure P6.3 is used, where $G(s) = G_c(s)P(s)$. The following transfer functions have been derived for $G(s)$ (*Tsang*, *2009*):

$$P(s) = \frac{R_1R_2C_1C_2s^2 + (R_1C_1 + R_2C_1 + R_2C_2)s + 1}{C_1(1 + R_2C_2)s}$$

$$G_c(s) = K_p + \frac{K_I}{s}$$

If $R_1 = 0.15\,\Omega$; $R_2 = 0.44\,\Omega$; $C_1 = 7200$ F; and $C_2 = 170$ F use the Routh-Hurwitz criteria to find the range of positive K_P and K_I for which the system is closed-loop stable.

65. Figure P6.19 is a simplified and linearized block diagram of a cascade control system, which is used to control water level in a steam generator of a nuclear power plant (*Wang*, *2009,*).

In this system, the level controller, $G_{LC}(s)$, is the *master* controller and the feed-water flow controller, $G_{FC}(s)$ is the *slave* controller. Using mass balance equations, the water level would ordinarily be regarded as a simple integration process of water flow. In a steam generator, however, steam flow rate and the cooling effect of feed-water change the dynamics of that process. Taking the latter into account and ignoring the much-less pronounced impact of changes in steam flow rate, a first-order lag plus time delay is introduced into the transfer

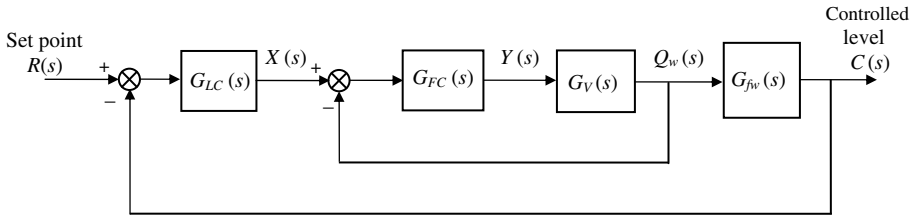


FIGURE P6.19

function, $G_{fw}(s)$, relating the controlled level, $C(s)$, to feed-water flow rate, $Q_w(s)$ as follows:

$$G_{fw}(s) = \frac{C(s)}{Q_w(s)} = \frac{K_1 e^{-\tau_1 S}}{s(T_1 s + 1)} = \frac{2e^{-2S}}{s(25s+1)} \approx \frac{2}{s(25s+1)(2s^2+2s+1)}$$

where $K_1 = 2$ is the process gain, $\tau_1 = 2$ is the pure time delay, and $T_1 = 25$ is the steam generator's time constant. (The expression $e^{-\tau_1 s}$ represents a time delay. This function can be represented by what is known as a *Pade approximation*. This approximation can take on many increasingly complicated forms, depending upon the degree of accuracy required. Here we use the Pade approximation, $e^{-x} \approx \dfrac{1}{1 + x + \dfrac{x^2}{2!}}$, and specific numerical values for the considered steam generator.)

The dynamic characteristics of the control valve are approximated by the transfer function: $G_v(s) = \dfrac{Q_w(s)}{Y(s)} = \dfrac{K_v}{T_v s + 1} = \dfrac{1}{3s+1}$, where K_v is the valve gain and T_v is its time constant.

Given that: $G_{FC}(s) = K_{P_{FC}} + K_{D_{FC}} s = 0.5 + 2s$ and $G_{LC}(s) = K_{P_{LC}} + K_{D_{LC}} s = 0.5 + K s$, use the Routh-Hurwitz criterion to find the range of the level controller's derivative gain, $K_{D_{LC}} = K > 0$, that will keep the system stable.

66. Look-ahead information can be used to automatically steer a bicycle in a closed-loop configuration. A line is drawn in the middle of the lane to be followed, and an arbitrary point is chosen in the vehicle's longitudinal axis. A look-ahead offset is calculated by measuring the distance between the look-ahead point and the reference line and is used by the system to correct the vehicle's trajectory. A linearized model of a particular bicycle traveling on a straight-line path at a fixed longitudinal speed is

$$\begin{bmatrix} \dot{V} \\ \dot{r} \\ \dot{\psi} \\ \dot{Y}_g \end{bmatrix} = \begin{bmatrix} -11.7 & 6.8 & 61.6K & 7.7K \\ -3.5 & -24 & -66.9K & 8.4K \\ 0 & 1 & 0 & 0 \\ 1 & 0 & -10 & 0 \end{bmatrix} \begin{bmatrix} V \\ r \\ \psi \\ Y_g \end{bmatrix}$$

In this model, V = bicycle's lateral velocity, r = bicycle's yaw velocity, ψ = bicycle's yaw acceleration, and Y_g = bicycle's center of gravity coordinate on the y-axis. K is a controller parameter to be chosen by the designer (*Özgüner, 1995*). Use the Routh-Hurwitz citerion to find the range of K for which the system is closed-loop stable.

PROGRESSIVE ANALYSIS AND DESIGN PROBLEMS

67. **High-speed rail pantograph.** Problem 21 in Chapter 1 discusses active control of a pantograph mechanism for high-speed rail systems. In Problem 79(a), Chapter 5, you found the block diagram for the active pantograph control system. Using your solution for Problem 79(a) in Chapter 5 and the Routh-Hurwitz criterion, find the range of controller gain, K, that will keep the system stable (*O'Connor, 1997*).

68. **Control of HIV/AIDS.** The HIV infection linearized model developed in Problem 82, Chapter 4, can be shown to have the transfer function

$$P(s) = \frac{Y(s)}{U_1(s)} = \frac{-520s - 10.3844}{s^3 + 2.6817s^2 + 0.11s + 0.0126}$$

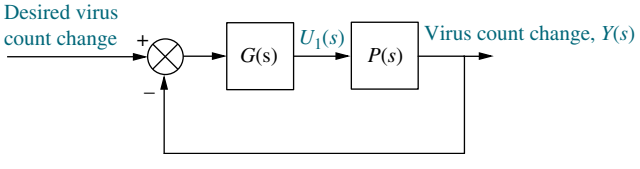


FIGURE P6.20

It is desired to develop a policy for drug delivery to maintain the virus count at prescribed levels. For the purpose of obtaining an appropriate $u_1(t)$, feedback will be used as shown in Figure P6.20 (*Craig, 2004*).

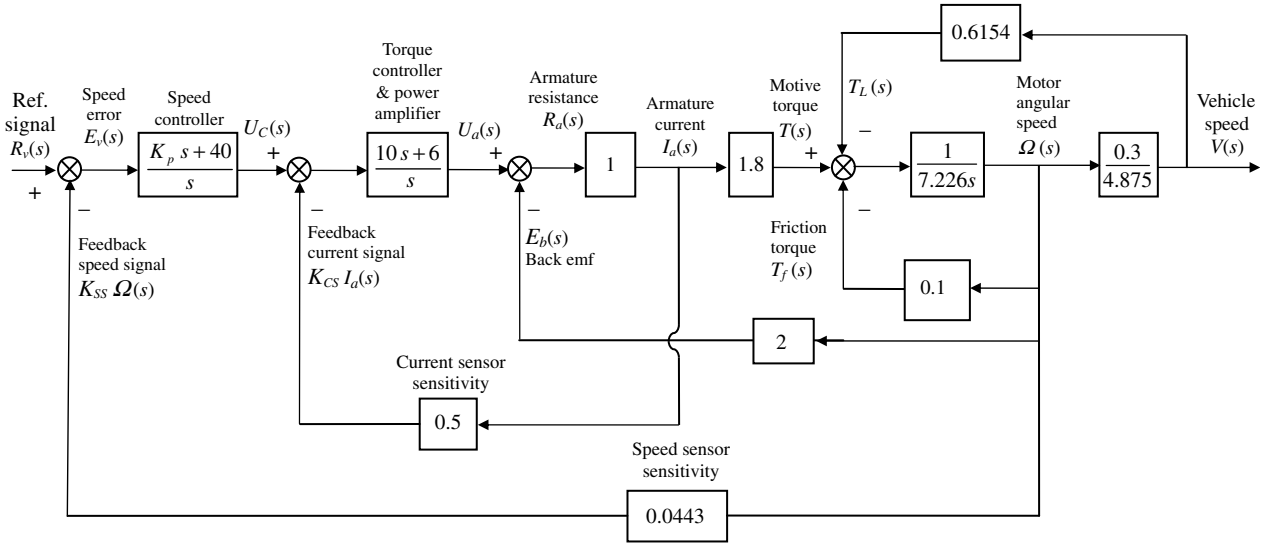


FIGURE P6.21

As a first approach, consider $G(s) = K$, a constant to be selected. Use the Routh-Hurwitz criteria to find the range of K for which the system is closed-loop stable.

69. Hybrid vehicle. Figure P6.21 shows the HEV system presented in Chapter 5, where parameter values have been substituted. It is assumed here that the speed controller has a proportional gain, K_p, to be adjusted. Use the Routh-Hurwitz stability method to find the range of positive K_p for which the system is closed-loop stable (*Graebe, 1995*).

Cyber Exploration Laboratory

Experiment 6.1

Objectives To verify the effect of pole location upon stability. To verify the effect upon stability of loop gain in a negative feedback system.

Minimum Required Software Packages MATLAB, Simulink, and the Control System Toolbox

Prelab

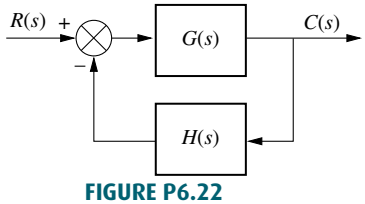


FIGURE P6.22

1. Find the equivalent transfer function of the negative feedback system of Figure P6.22 if

$$G(s) = \frac{K}{s(s+2)^2} \text{ and } H(s) = 1$$

2. For the system of Prelab 1, find two values of gain that will yield closed-loop, overdamped, second-order poles. Repeat for underdamped poles.

3. For the system of Prelab 1, find the value of gain, K, that will make the system critically damped.

4. For the system of Prelab 1, find the value of gain, K, that will make the system marginally stable. Also, find the frequency of oscillation at that value of K that makes the system marginally stable.
5. For each of Prelab 2 through 4, plot on one graph the pole locations for each case and write the corresponding value of gain, K, at each pole.

Lab

1. Using Simulink, set up the negative feedback system of Prelab 1. Plot the step response of the system at each value of gain calculated to yield overdamped, underdamped, critically damped, and marginally stable responses.
2. Plot the step responses for two values of gain, K, above that calculated to yield marginal stability.
3. At the output of the negative feedback system, cascade the transfer function

$$G_1(s) = \frac{1}{s^2 + 4}$$

Set the gain, K, at a value below that calculated for marginal stability and plot the step response. Repeat for K calculated to yield marginal stability.

Postlab

1. From your plots, discuss the conditions that lead to unstable responses.
2. Discuss the effect of gain upon the nature of the step response of a closed-loop system.

Experiment 6.2

Objective To use the LabVIEW Control Design and Simulation Module for stability analysis.

Minimum Required Software Package LabVIEW with the Control Design and Simulation Module

Prelab Select six transfer functions of various orders and use Routh-Hurwitz to determine their stability.

Lab Create a LabVIEW VI that receives the order and the coefficients of the characteristic equation and outputs the location of the poles and information regarding stability.

Postlab Verify the stability of the systems from your Prelab.

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